

Time-varying mesh stiffness calculation of cracked spur gears



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ABSTRACT

Considering the misalignment of gear root circle and base circle and accurate transition curve, an improved mesh stiffness model for a healthy gear pair is proposed and validated by the finite element method (FEM). Based on the improved method, three mesh stiffness calculation methods (method 1: straight lines for crack path and limiting line; method 2: straight line for crack path and parabolic curve for limiting line proposed in Ref. [1]; method 3: parabolic curves for crack path and limiting line) for cracked gear pair are presented and compared with FEM. The results show that there is a significant difference between method 1 and FEM under large crack condition and the results of methods 2 and 3 are quite close to FEM result, which also shows that the parabolic curve as a limiting line is appropriate. Mesh stiffness of method 2 is very close to that of method 3, which also shows that it is acceptable to assume the crack path to be a straight line.

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1. Introduction

Gear tooth crack may appear under severe operating conditions, such as excessive service load, inappropriate operating conditions, or simply fatigue [1,2]. The gear mesh stiffness will reduce, which will increase the vibration and noise of gears when the tooth crack appears. Both the FEM and the analytical method (AM) have been utilized to calculate the TVMS of gears. The two methods have advantages and disadvantages; in the case of FEM, the calculation result is more close to actual situation because it can consider accurate deformations of involute, transition curve and fillet foundation. For AM, it can offer a simple and effective way to calculate the TVMS; however, FEM is complicated and time consuming compared with AM.

Many researchers have developed analytical methods to study the TVMS of healthy or cracked gears. Considering Hertzian, bending and axial compressive energy, Yang and Lin [3] calculated the TVMS of gears by the potential energy principle. Based on the potential energy method proposed by Yang and Lin [3], Tian [4] discussed the effects of a chipped tooth, a cracked tooth and a broken tooth on TVMS by considering the shear energy and compared the vibration acceleration responses of the gear system under different fault types. The effects of the crack propagation and the rim thickness on crack paths have been performed by many researchers [5–10]. The results of these studies can be used to simplify crack propagation paths, which will provide information to calculate TVMS under different crack levels. On the basis of Refs. [4,6,7], Wu et al. [11] developed an analytical model for calculating the TVMS under a cracked tooth at different levels and studied the effects of tooth crack growth on the vibration response of a one-stage gearbox with spur gears by computer simulation. Pandya and Parey [12] carried out the crack propagation path studies of high contact ratio spur gear with tooth root crack for two gear parameters viz. backup ratio and pressure angle. On the basis of the predicted crack paths, they estimated the mesh stiffness variation by a total potential energy model presented in Ref. [11]. Based on the corrected tooth foundation

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deformation proposed by Sainsot et al. [13], Chaari et al. [14] presented an original analytical model of tooth crack, quantified the gear mesh stiffness reduction due to spur gear tooth crack and verified the results obtained analytically by comparing with the results obtained by FEM. Assuming the gear body as a cantilever beam in the half plane, Zhou et al. [15] developed a modified mathematical model for simulating gear crack from root with linear growth path in a pinion by considering the deformation of gear-body. Considering both the tooth crack propagations along tooth width and crack depth, Chen and Shao [16,17] proposed an analytical model to investigate the effect of gear tooth crack on the TVMS. By this analytical formulation, the mesh stiffness of a spur gear pair with different crack lengths and depths can be obtained. By assuming parabolic curves instead of straight lines to deal with the tooth thickness reduction, Mohammed et al. [1] presented a new method to calculate the TVMS of the gear pair for a propagating crack in the tooth root and investigated the influence of gear mesh stiffness on the vibration-based fault detection indicators, the RMS, Kurtosis and the crest factor. In consideration that root circle and base circle are misaligned exactly, Wan et al. [18] developed a modified method to obtain the TVMS accurately by judging the number of gear teeth, established a multi-degree of freedom dynamic model for gearbox system with cracked tooth and analyzed the dynamic response of the system by Runge–Kutta method. Liang et al. [19] adopted a modified cantilever beam model to represent the external gear tooth, derived the analytical equations of the bending, shear and axial compressive stiffness by considering the gear tooth starting from the root circle and developed a crack propagation model to quantify the mesh stiffness reduction when a crack occurs in the sun gear or the planet gear.

Many researchers also analyzed the TVMS of gears based on both 2D and 3D finite element (FE) models. Chaari et al. [14] adopted a 3D FE model to analyze the TVMS of gears. In order to simplify the computation of tooth deflection, only one tooth was considered and FE contact between pinion and wheel tooth pairs is not considered. By applying a linear distributed force to the tooth flank normal to the involute profile and along the line of action at the appropriate nodes which simulates the action of the meshing tooth of the wheel, they calculated the deflection of the tooth along the direction of the force and obtained the TVMS of cracked gears. By the similar method provided in Ref. [14], Mohammed et al. [1] established a 2D FE model (plane stress problem) to calculate the TVMS of spur gears. By subdividing the gear tooth deformations into a global and a local type, Fernandez del Rincon et al. [20] calculated these deformations by a FE model and a nonlinear Hertzian-based formulation, respectively. This procedure enables a reduced computational effort, in contrast to conventional FE models with contact elements. For simulating the contact process of healthy or cracked gear teeth accurately, many researchers also calculated TVMS by defining contact elements. Lin et al. [21] presented an approach for mesh generation of gear drives at any meshing position, developed an automatic modeling program to simulate 3D contact/impact problem of gear teeth and obtained tooth load distribution and mesh stiffness results are derived under static loading during meshing process. Based on a plane strain FE model, Ma et al. [22] calculated TVMS of the tip relieved involute spur gear pairs without considering the extension of the meshing period of a tooth pair due to dynamic overloads. Based on the software package ANSYS, Wang [23] presented a numerical approach to predict the TVMS of spur gears by establishing contact elements.

In the above-mentioned researches, the gear tooth is mostly modeled as a nonuniform cantilever beam on base circle when the TVMS of spur gears is solved based on the energy method. However, root circle and base circle are misaligned exactly, so the solution error of TVMS will appear. Moreover, the crack in the tooth root will propagate through the tooth in a gear with a high backup ratio and the path is similar to a slight curve (see Fig. 1). However, most researches assumed the crack path as a straight line which also lead to a slight error for estimating TVMS of gears.

Aiming at these problems, an improved mesh stiffness model is proposed, in which the gear tooth is simplified as a cantilever beam on the root circle and the accurate transition curve is also considered. Moreover, a parabolic curve instead of straight line to deal with the tooth thickness reduction [1] and a parabolic curve instead of straight line to simulate crack propagation path are modified in the present paper. In this study, different crack levels are also examined with three methods (method 1: straight lines for crack path and limiting line; method 2: straight line for crack path and parabolic curve for limiting line; method 3: parabolic curves for crack path and limiting line). Emphasis is given to the effects of different analytical methods on the TVMS of the gears by comparing with FEM.

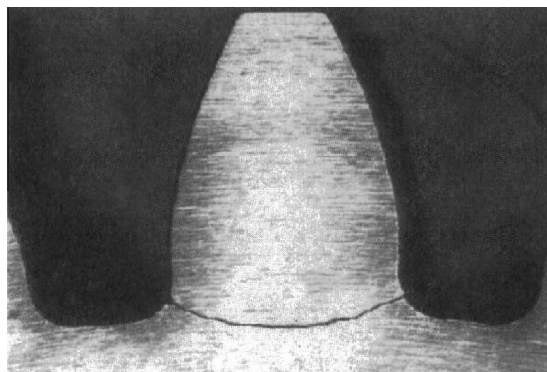


Fig. 1. Measured gear crack propagation path in Ref. [5].

2. Mesh stiffness calculation of a healthy spur gear pair

2.1. An improved mesh stiffness model for a healthy spur gear pair

In many published literatures, the gear tooth is generally modeled as a nonuniform cantilever beam on base circle. However, root circle and base circle are misaligned exactly, especially when the number of teeth is larger or smaller. Assuming that the radiuses of the base circle and root circle of the gear are r_b and r_f , and their equations can be expressed as:

$$r_b = \frac{1}{2}mN \cos \alpha, \quad r_f = \frac{1}{2}mN - (h_a^* + c^*)m, \quad (1)$$

where m is the module, N the number of teeth, α the pressure angle of the gear pitch circle, h_a^* the addendum coefficient and c^* the tip clearance coefficient. If the gear is a standard spur gear with $h_a^* = 1$, $c^* = 0.25$ and $\alpha = 20^\circ$, when $N = 41.45$, the root circle will overlap with the base circle. So the root circle is bigger than the base circle (see Fig. 2a) if the teeth number N is more than 41. Reversely, the root circle is smaller than the base circle (see Fig. 2b) if the teeth number N is less than or equal to 41.

In this section, an improved mesh stiffness model is presented, in which the misalignment between the root circle and base circle is considered and **the gear tooth is simplified as a cantilever beam on the root circle**. In addition, the accurate transition curve is also considered in the improved method.

Tooth profile curve can be divided into four parts, **addendum curve AB**, **involute curve BC**, **transition curve CD** and **dedendum curve DE** (see Fig. 3). Equations of involute curve are expressed as follows:

$$\begin{cases} x = r_i \sin \varphi \\ y = r_i \cos \varphi \end{cases}, \quad (2)$$

$$\varphi = \frac{\pi}{2N} - (\text{inv} \alpha_i - \text{inv} \alpha), \quad (3)$$

where $r_i = r_b / \cos \alpha_i$; α_i ($\alpha_c \leq \alpha_i \leq \alpha_a$) is the pressure angle of the arbitrary point at the involute curve, in which $\alpha_c = \arccos(r_b/r_c)$, $\alpha_a = \arccos(r_b/r_a)$ is the pressure angle of the involute starting point and the addendum circle, respectively;

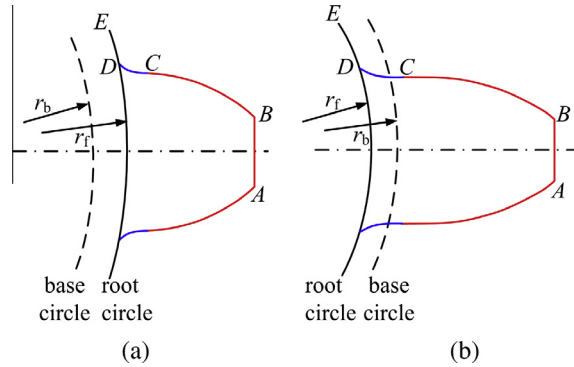


Fig. 2. Misalignment between base circle and root circle: (a) $N > 41$ (b) $N \leq 41$.

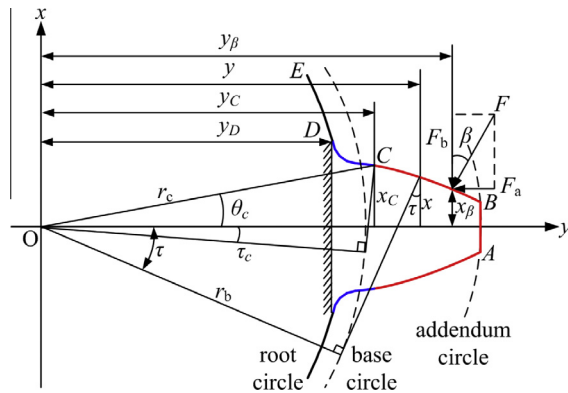


Fig. 3. Geometric model of the gear profile.

$r_c = \sqrt{(r_b \tan \alpha - h_a^* m / \sin \alpha)^2 + r_b^2}$ is the radius of the involute starting point circle; $\text{inv} \alpha_i$ is the involute function, $\text{inv} \alpha_i = \tan \alpha_i - \alpha_i$, $\text{inv} \alpha = \tan \alpha - \alpha$.

The transition curve is cut out by the cutter tip. From the perspective of the machining process of a gear tooth, transition curve is the tooth profile between the involute starting point and the root circle, and no matter how many teeth the gear has, the transitional part will always exist. That is to say, no matter the root circle is bigger or smaller than the base circle, the transition curve will always exist. The shape of the transition curve is directly dependent on the shape of the cutter tip. When the shape of the cutter tip is ordinary fillet, the transition curve equations are [24]:

$$\begin{cases} x = r \times \sin(\Phi) - (a_1 / \sin \gamma + r_\rho) \times \cos(\gamma - \Phi) \\ y = r \times \cos(\Phi) - (a_1 / \sin \gamma + r_\rho) \times \sin(\gamma - \Phi) \end{cases} (\alpha \leq \gamma \leq \pi/2), \quad (4)$$

where r is the radius of the pitch circle, $\Phi = (a_1 / \tan \gamma + b_1) / r$, $r_\rho = c^* m / (1 - \sin \alpha)$, $a_1 = (h_a^* + c^*) \times m - r_\rho$, and $b_1 = \pi m / 4 + h_a^* m \tan \alpha + r_\rho \cos \alpha$.

Based on elastic mechanics, the relationships between Hertzian contact stiffness k_h , bending stiffness k_b , shear stiffness k_s , axial compressive stiffness k_a , fillet foundation stiffness k_f and Hertzian energy U_h , bending energy U_b , shear energy U_s , axial compressive energy U_a , fillet foundation energy U_f are obtained, which are written as:

$$U_h = \frac{F^2}{2k_h}, U_b = \frac{F^2}{2k_b}, U_s = \frac{F^2}{2k_s}, U_a = \frac{F^2}{2k_a}, U_f = \frac{F^2}{2k_f}, \quad (5)$$

where F is the total acting force on the contact teeth.

The summation of Hertzian, bending, shear, axial compressive and fillet foundation energies constitutes the total potential energy U stored in a single pair of meshing teeth, which can be expressed as:

$$\begin{aligned} U &= \frac{F^2}{2k} = U_h + U_{b1} + U_{s1} + U_{a1} + U_{f1} + U_{b2} + U_{s2} + U_{a2} + U_{f2} \\ &= \frac{F^2}{2} \left(\frac{1}{k_h} + \frac{1}{k_{b1}} + \frac{1}{k_{s1}} + \frac{1}{k_{a1}} + \frac{1}{k_{f1}} + \frac{1}{k_{b2}} + \frac{1}{k_{s2}} + \frac{1}{k_{a2}} + \frac{1}{k_{f2}} \right), \end{aligned} \quad (6)$$

where k represents the total effective mesh stiffness of a pair of meshing teeth, subscripts 1 and 2 denote pinion and gear, respectively. According to Eq. (6), single-tooth-pair mesh stiffness can be given as:

$$k = \frac{1}{\frac{1}{k_h} + \frac{1}{k_{b1}} + \frac{1}{k_{s1}} + \frac{1}{k_{a1}} + \frac{1}{k_{f1}} + \frac{1}{k_{b2}} + \frac{1}{k_{s2}} + \frac{1}{k_{a2}} + \frac{1}{k_{f2}}}. \quad (7)$$

Consequently, the total TVMS of the gear pair in the mesh cycle with contact ratio between 1 and 2 can be calculated as:

$$k_{12} = \sum_{i=1}^j k_i, \quad (j = 1, 2), \quad (8)$$

where k_{12} is the total TVMS of the gear pair in the mesh cycle, j denotes the number of meshing tooth pair at the same time.

Hertzian contact stiffness and fillet foundation stiffness [14,16] can be found in Eqs. (9) and (10).

$$k_h = \frac{\pi EL}{4(1 - \nu^2)}, \quad (9)$$

where E , L and ν represent Young's modulus, tooth width and Poisson's ratio, respectively.

$$\frac{1}{k_f} = \frac{\cos^2 \beta}{EL} \left\{ L^* \left(\frac{u_f}{S_f} \right)^2 + M^* \left(\frac{u_f}{S_f} \right) + P^* (1 + Q^* \tan^2 \beta) \right\}, \quad (10)$$

where β is the operating pressure angle (see Fig. 3), parameters u_f , S_f , L^* , M^* , P^* and Q^* can be found in Ref. [13].

Based on beam theory, axial compressive, bending and shear energies of gear tooth comprised of involute and transitional part can be calculated by:

$$U_a = \frac{F^2}{2k_a} = \int_{y_D}^{y_C} \frac{F_a^2}{2EA_{y1}} dy_1 + \int_{y_C}^{y_\beta} \frac{F_a^2}{2EA_{y2}} dy_2, \quad (11)$$

$$U_b = \frac{F^2}{2k_b} = \int_{y_D}^{y_C} \frac{M_1^2}{2EI_{y1}} dy_1 + \int_{y_C}^{y_\beta} \frac{M_2^2}{2EI_{y2}} dy_2, \quad (12)$$

$$U_s = \frac{F^2}{2k_s} = \int_{y_D}^{y_C} \frac{1.2F_b^2}{2GA_{y1}} dy_1 + \int_{y_C}^{y_\beta} \frac{1.2F_b^2}{2GA_{y2}} dy_2, \quad (13)$$

where $F_b = F \cos \beta$, $F_a = F \sin \beta$, x_β is the distance between the contact point and the central line of the tooth, y_β is the distance between the contact point and original point in the horizontal direction, $G = \frac{E}{2(1+\nu)}$ is the shear modulus, y_1, y_2 represent the horizontal coordinates of arbitrary point at transition curve and involute curve, y_C, y_D denote starting point and ending point of the transition curve, $I_{y1}, A_{y1}, I_{y2}, A_{y2}$ represent the area moment of inertia and the cross-sectional area, respectively; torques $M_1 = F_b(y_\beta - y_1) - F_a x_\beta$ and $M_2 = F_b(y_\beta - y_2) - F_a x_\beta$ denote the bending effect of F_b and F_a on the part of transition and involute curves.

For convenience, angular displacement is adopted in the calculation. According to the characteristics of involute and transition curve, $x_\beta, y_\beta, x_1, y_1, x_2, y_2, I_{y1}, I_{y2}, A_{y1}$ and A_{y2} can be expressed as the functions of angles γ, β or τ (see Appendix A), and then axial compressive, bending and shear stiffness can be written as follows based on Eqs. (11)–(13):

$$\frac{1}{k_a} = \int_{\frac{\pi}{2}}^{\alpha} \frac{\sin^2 \beta}{EA_{y1}} \frac{dy_1}{d\gamma} d\gamma + \int_{\tau_c}^{\beta} \frac{\sin^2 \beta}{EA_{y2}} \frac{dy_2}{d\tau} d\tau, \quad (14)$$

$$\frac{1}{k_b} = \int_{\frac{\pi}{2}}^{\alpha} \frac{[\cos \beta (y_\beta - y_1) - x_\beta \sin \beta]^2}{EI_{y1}} \frac{dy_1}{d\gamma} d\gamma + \int_{\tau_c}^{\beta} \frac{[\cos \beta (y_\beta - y_2) - x_\beta \sin \beta]^2}{EI_{y2}} \frac{dy_2}{d\tau} d\tau, \quad (15)$$

$$\frac{1}{k_s} = \int_{\frac{\pi}{2}}^{\alpha} \frac{1.2 \cos^2 \beta}{GA_{y1}} \frac{dy_1}{d\gamma} d\gamma + \int_{\tau_c}^{\beta} \frac{1.2 \cos^2 \beta}{GA_{y2}} \frac{dy_2}{d\tau} d\tau, \quad (16)$$

where $\frac{dy_1}{d\gamma}$ and $\frac{dy_2}{d\tau}$ are shown in Appendix A.

2.2. Verification for improved mesh stiffness model by FE model

Two gear pairs are selected to verify the validity of the improved mesh stiffness model and their parameters are listed in Table 1. TVMS based on traditional method (TM) in which the gear tooth is modeled as a nonuniform cantilever beam on base circle, improved method (IM) in which the gear tooth is simplified as a cantilever beam on the root circle and FEM with precise profile curve introduced in Ref. [14] are displayed in Fig. 4. Obviously, the results from IM and FEM show good agreement, while TM shows great difference from FEM. Thus it can be seen that IM is more accurate to calculate the TVMS than TM compared with the result of FEM.

Table 1
Parameters of spur gear pairs.

Parameters	Gear pair 1		Gear pair 2	
	Pinion	Gear	Pinion	Gear
Number of teeth N	22	22	62	62
Young's modulus E (GPa)	206	206	206	206
Poisson's ratio ν	0.3	0.3	0.3	0.3
Module m (mm)	3	3	3	3
Addendum coefficient h_a^*	1	1	1	1
Tip clearance coefficient c^*	0.25	0.25	0.25	0.25
Tooth width L (mm)	20	20	20	20
Pressure angle α (°)	20	20	20	20
Hub bore radius r_{int} (mm)	11.7	11.7	35.7	35.7

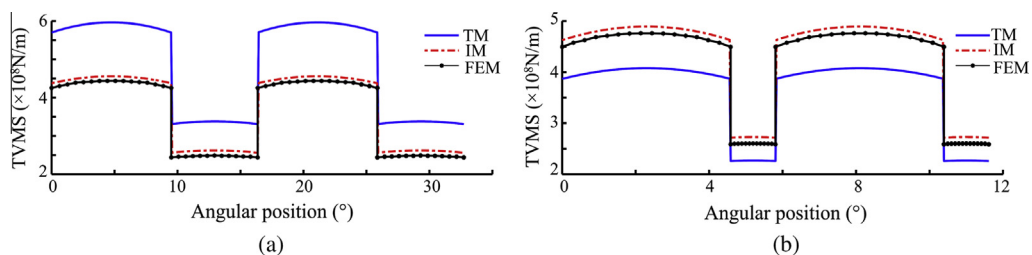


Fig. 4. TVMS based on different methods: (a) gear pair 1, (b) gear pair 2.

3. TVMS calculation using different methods

3.1. TVMS calculation using method 1

In method 1 the crack path is assumed to be a straight line. The crack starts at the root of the driven gear and then propagates, as is shown in Fig. 5. The geometrical parameters of the crack (q , v , ψ) are presented in Fig. 5, in which q denotes the crack depth, v the crack propagation direction, ψ the crack initial position (in this paper $\psi = 35^\circ$).

Assuming that the crack extends along the whole tooth width with a uniform crack depth distribution, the tooth with this crack is still considered as a cantilevered beam, the curve of the tooth profile remains perfect and the foundation stiffness is not affected. In this case, based on Eq. (9), the Hertzian contact stiffness will still be a constant since the work surface of the tooth has no defect and the width of the effective work surface will always be a constant L . For the axial compressive stiffness, it will be considered the same with that under the perfect condition in that the crack part can still bear the axial compressive force as if no crack exists. Therefore, for the gear with a cracked tooth, Hertzian, fillet foundation and axial compressive stiffness can still be calculated according to Eqs. (9), (10), and (14), respectively. However, the bending and shear stiffnesses will change due to the influence of the crack. Therefore, based on Eq. (7) single-tooth-pair mesh stiffness for a cracked gear pair can be calculated as:

$$k = \frac{1}{\frac{1}{k_h} + \frac{1}{k_{b1}} + \frac{1}{k_{s1}} + \frac{1}{k_{a1}} + \frac{1}{k_{f1}} + \frac{1}{k_{b_crack}} + \frac{1}{k_{s_crack}} + \frac{1}{k_{a2}} + \frac{1}{k_{f2}}}, \quad (17)$$

where k_{b_crack} and k_{s_crack} denote bending and shear stiffnesses when a crack is introduced.

When the tooth crack is present, and $q = q_1 \leq q_{1\max}$ (see Fig. 5a), the effective area moment of inertia and area of the cross section at the position of y can be calculated as:

$$I = \begin{cases} \frac{1}{12} (h_q + x)^3 L, & y_p \leq y \leq y_G \\ \frac{1}{12} (2x)^3 L = \frac{2}{3} x^3 L, & y < y_p \text{ or } y > y_G \end{cases}, \quad (18)$$

$$A = \begin{cases} (h_q + x)L, & y_p \leq y \leq y_G \\ 2xL, & y < y_p \text{ or } y > y_G \end{cases}, \quad (19)$$

where h_q is the distance from the root of the crack to the central line of the tooth, which corresponds to point G on the tooth profile, and h_q can be calculated by:

$$h_q = x_Q - q_1 \sin v. \quad (20)$$

When the crack propagates to the central line of the tooth, q_1 reaches its maximum value $q_{1\max} = x_Q / \sin v$. Then the crack will change direction to q_2 , which is assumed to be exactly symmetric with q_1 , and in this stage $q = q_{1\max} + q_2$ (see Fig. 5b). When the crack propagates along q_2 , the effective area moment of inertia and area of the cross section at the position of y can be calculated as:

$$I = \begin{cases} \frac{2}{3} x^3 L, & y < y_{\max} \\ \frac{1}{12} \left[x - \frac{(y - y_{\max}) h_q}{y_p - y_{\max}} \right]^3 L, & y_{\max} \leq y < y_p \\ \frac{1}{12} (x - h_q)^3 L, & y_p \leq y \leq y_G \\ 0, & y > y_G \end{cases}, \quad (21)$$

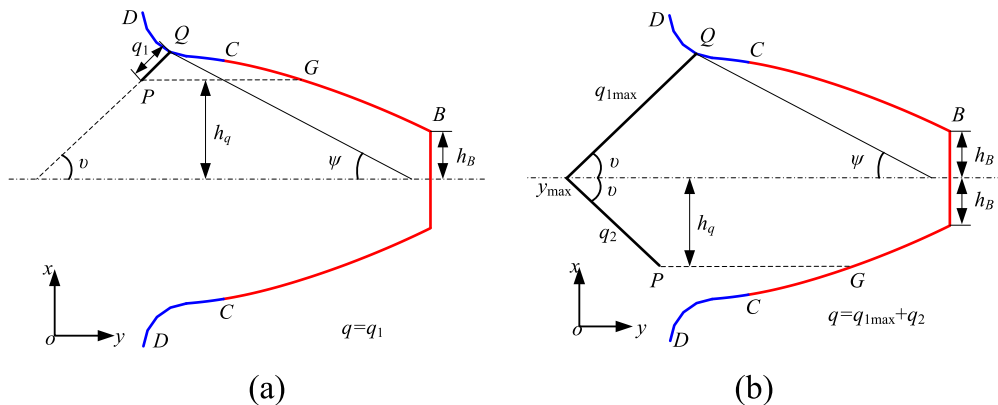


Fig. 5. Schematic of crack propagation for method 1: (a) $q \leq q_{1\max}$, (b) $q > q_{1\max}$.

$$A = \begin{cases} 2xL, & y < y_{\max} \\ \left[x - \frac{(y-y_{\max})h_q}{y_p-y_{\max}} \right] L, & y_{\max} \leq y < y_p \\ (x-h_q)L, & y_p \leq y \leq y_G \\ 0, & y > y_G \end{cases}, \quad (22)$$

where $y_{\max}$ denotes the leftmost position of crack propagation (shown in Fig. 5b), and $h_q = q_2 \sin \nu$.

Based on Eqs. (15) and (16), the bending and shear stiffnesses of the cracked gears can be calculated. The schematic of the calculation process for method 1 is presented in Fig. 6.

In this paper, the parameters of the studied spur gear pair are shown in Table 2. In order to analyze the influences of crack depth and crack propagation direction on TVMS, 10 crack cases are considered, as are shown in Table 3. The crack depth is expressed as a percentage which is defined as the ratio of actual crack depth with theoretical through-tooth crack size.

TVMS under the 10 cases are displayed in Fig. 7. In the figure, distinct reduction of stiffness can be observed when a tooth crack is introduced, and the maximum stiffness reduction during the mesh process for a fixed crack appears where the cracked tooth is just going to engagement. This is because the relative bigger flexibility of the tooth at the addendum circle compared to that at the position on the tooth profile which is closer to dedendum circle. In addition, the stiffness reduction increases with the growth of crack depth. Obviously when the crack depth percentage is the same, h_q has the same value, so if the transition part of the tooth is not considered, the mesh stiffness under different propagation direction angles should be completely the same. But the stiffness in Fig. 7 increases slightly with the increase of crack propagation direction angle ν , this is because a crack with a pretty large ν has little effect on the rigidity of the transition part. So when the transition part of the gear tooth is considered, not only the TVMS of the healthy gear pair will change (see Fig. 4), but the TVMS of the cracked gear pair will vary with the change of crack propagation direction angles.

3.2. TVMS calculation using method 2

Instead of resorting to a straight line (see straight line PG in Fig. 5) as a limiting line for reducing the tooth thickness in method 1, a parabolic curve between the crack tip and the tooth profile tip is chosen as the limiting line in method 2 (see Fig. 8), which is proposed by Mohammed et al. [1] and the crack propagation path is assumed to be the same as that in method 1.

Similarly, once k_{p_crack} and k_{s_crack} are determined, the single-tooth-pair mesh stiffness for cracked gears can be obtained by Eq. (17). Effective area moment of inertia and area of the cross section at the position of y in method 2 can be calculated as follows:

When $q = q_1 \leq q_{1\max}$ (see Fig. 8a),

$$I = \begin{cases} \frac{1}{12} (2x)^3 L = \frac{2}{3} x^3 L, & y \leq y_p \\ \frac{1}{12} \left[\frac{x_B - x_p}{(y_B - y_p)^2} (y - y_p)^2 + x_p + x \right]^3 L, & y > y_p \end{cases}, \quad (23)$$

$$A = \begin{cases} 2xL, & y \leq y_p \\ \left[\frac{x_B - x_p}{(y_B - y_p)^2} (y - y_p)^2 + x_p + x \right] L, & y > y_p \end{cases}. \quad (24)$$

When $q = q_{1\max} + q_2 > q_{1\max}$ (see Fig. 8b),

$$I = \begin{cases} \frac{2}{3} x^3 L, & y < y_{\max} \\ \frac{1}{12} \left[x - \frac{(y-y_{\max})x_p}{y_p-y_{\max}} \right]^3 L, & y_{\max} \leq y \leq y_p \\ \frac{1}{12} \left[\frac{x_B + x_p}{(y_B - y_p)^2} (y - y_p)^2 - x_p + x \right]^3 L, & y > y_p \end{cases}, \quad (25)$$

$$A = \begin{cases} 2xL, & y < y_{\max} \\ \left[x - \frac{(y-y_{\max})x_p}{y_p-y_{\max}} \right] L, & y_{\max} \leq y \leq y_p \\ \left[\frac{x_B + x_p}{(y_B - y_p)^2} (y - y_p)^2 - x_p + x \right] L, & y > y_p \end{cases}, \quad (26)$$

The schematic of the calculation process for method 2 is presented in Fig. 9.

Likewise, TVMS obtained by method 2 under 10 crack cases shown in Table 3 are displayed in Fig. 10 and the variation of TVMS is the same as that by method 1, except that the reduction of TVMS using method 2 is smaller than that from method 1 when the crack depth is larger and a detailed comparison will be done in Section 3.4.



Fig. 6. Schematic to obtain bending and shear stiffness for a cracked tooth for method 1.

Table 2
Parameters of the studied spur gear pair.

Parameters	Pinion (driving gear)	Gear (driven gear)
Number of teeth N	55	75
Young's modulus E (GPa)	212	212
Poisson's ratio ν	0.29	0.29
Module m (mm)	2	2
Addendum coefficient h_a^*	1	1
Tip clearance coefficient c^*	0.25	0.25
Tooth width L (mm)	20	20
Pressure angle α ($^\circ$)	20	20
Hub bore radius r_{int} (mm)	17.5	17.5

Table 3
Crack case data.

Crack case	Crack depth (%)	Crack propagation direction ν ($^\circ$)
1	Healthy gear	
2	20	75
3		85
4		90
5	40	75
6		85
7		90
8	60	75
9		85
10		90

3.3. TVMS calculation using method 3

In the previous methods the crack propagation path is assumed to be a straight line, but according to Ref. [5] the crack propagation path (see Fig. 1) is a slight curve extending from the tooth root in the gear with a high backup ratio. Lewicki [7] noted that crack propagation paths are smooth, continuous, and in most cases, rather straight with only a slight curvature. Pandya and Parey [9] also pointed out the crack grew following a curved trajectory. In order to reduce the error generated by straight line assumption, for method 3 a parabolic curve is adopted to simulate the crack propagation instead of straight line and the limiting line is the same as that in method 2 (see Fig. 11).

Assuming that the intersection angle ν corresponding to the leftmost point of crack propagation is known, the parabolic equation can be determined by crack initial point Q and angle ν , so effective area moment of inertia and area of the cross section at the position of y in method 3 can be calculated as follows.

When $q = q_1 \leq q_{1\text{max}}$ (see Fig. 11a), the expressions of the effective area moment of inertia and area of the cross section at the position of y are the same as those acquired from method 2 (see Eqs. (23) and (24)). But different from method 2, q_1 in method 3 stands for the arc length between crack initial point Q and crack tip P , not the straight-line distance between the two points and $q_{1\text{max}}$ in method 3 stands for the arc length between crack initial point Q and the leftmost point of crack propagation.

When $q = q_{1\text{max}} + q_2 > q_{1\text{max}}$ (see Fig. 11b),

$$I = \begin{cases} \frac{2}{3}x^3L, & y < y_{\text{max}} \\ \frac{1}{12} \left[x - \sqrt{\frac{(y-y_{\text{max}})x_Q^2}{y_Q-y_{\text{max}}}} \right]^3 L, & y_{\text{max}} \leq y \leq y_P, \\ \frac{1}{12} \left[\frac{x_B+x_P}{(y_B-y_P)^2} (y-y_P)^2 - x_P + x \right]^3 L, & y > y_P \end{cases} \quad (27)$$

$$A = \begin{cases} 2xL, & y < y_{\text{max}} \\ \left[x - \sqrt{\frac{(y-y_{\text{max}})x_Q^2}{y_Q-y_{\text{max}}}} \right] L, & y_{\text{max}} \leq y \leq y_P, \\ \left[\frac{x_B+x_P}{(y_B-y_P)^2} (y-y_P)^2 - x_P + x \right] L, & y > y_P \end{cases} \quad (28)$$

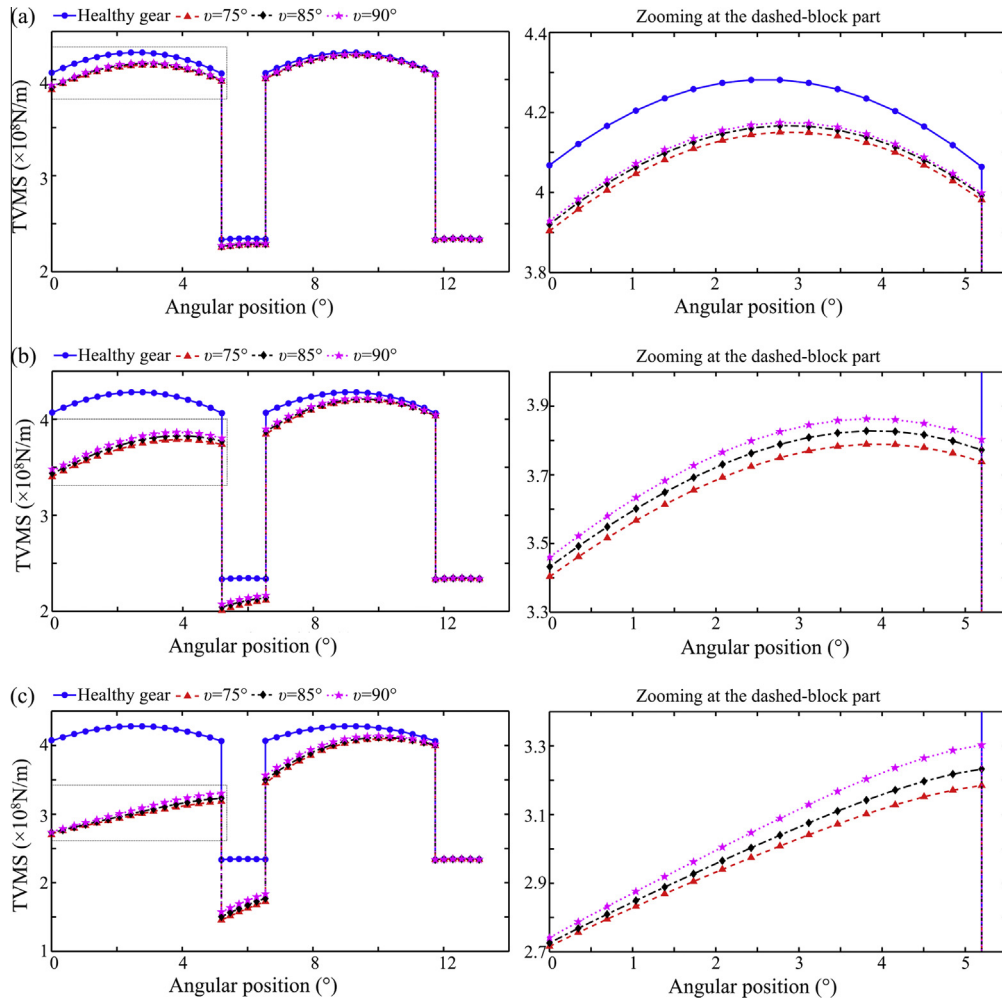


Fig. 7. TVMS of method 1 with different crack depths and propagation directions: (a) 20% crack, (b) 40% crack, (c) 60% crack.

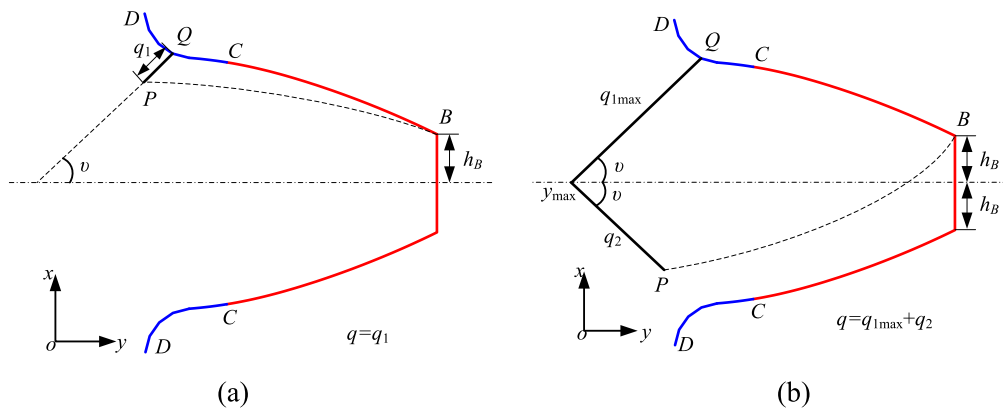


Fig. 8. Schematic of crack propagation for method 2: (a) $q \leq q_{1\max}$, (b) $q > q_{1\max}$.

The schematic of the calculation process in method 3 is basically the same with method 2 (see Fig. 9) except that B_3 , C_3 are replaced with $B_3 = \frac{1}{12} \left[x - \sqrt{\frac{(y-y_{\max})x_Q^2}{y_Q - y_{\max}}} \right]^3 L$ and $C_3 = \left[x - \sqrt{\frac{(y-y_{\max})x_Q^2}{y_Q - y_{\max}}} \right] L$.

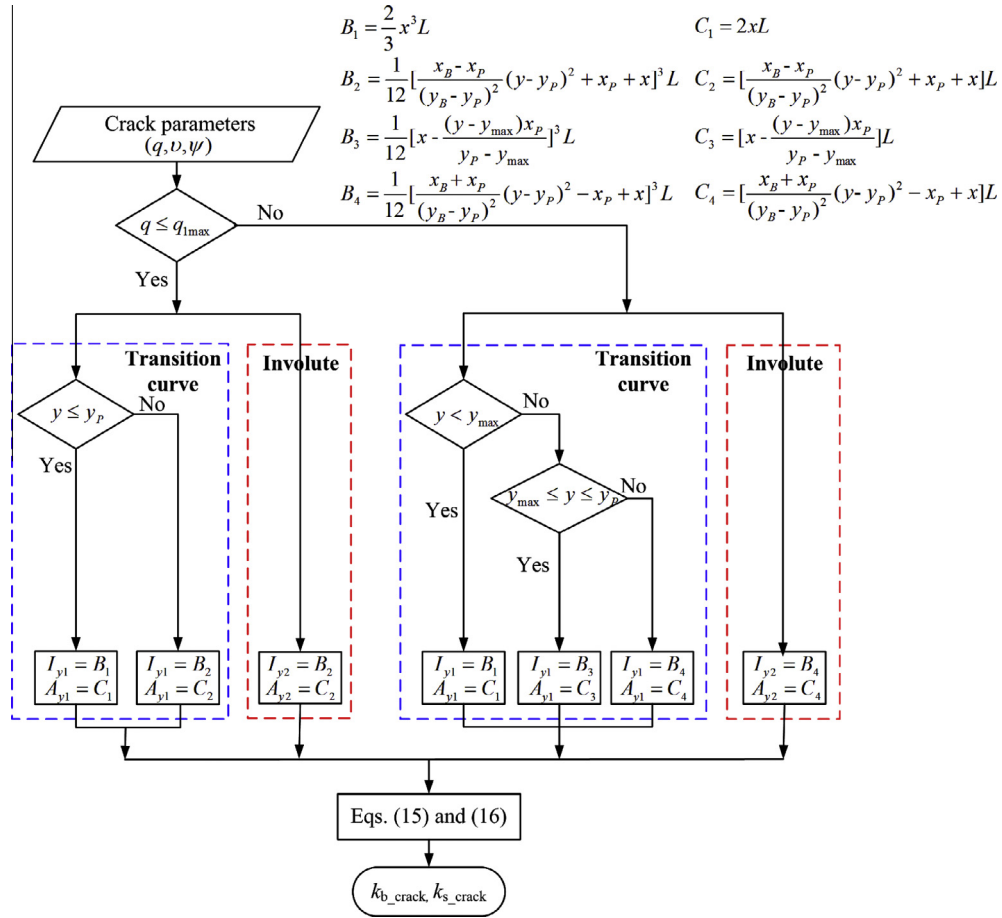


Fig. 9. Schematic to obtain bending and shear stiffness for a cracked tooth for method 2.

10 crack cases shown in Table 3 are also considered in method 3 in which v stands for the intersection angle corresponding to the leftmost point of crack propagation. TVMS obtained by method 3 under the 10 cases are displayed in Fig. 12 and the variation of TVMS is the same as that by method 2.

3.4. Comparison of TVMS using different methods

FEM is considered as an efficient and reliable tool for solving problems, and it is widely applied to calculate TVMS [1,14,20–23]. To reduce the computational time, analysis is performed by a 2D model with only one tooth (see Fig. 13), which is widely used and accepted in Refs. [1,14]. Cracks with different depths in tooth root are established by 2D singularity elements and a parabolic curve which is used in method 3 is adopted to simulate crack propagation. The load is considered to be in the plane of the gear body and uniformly distributed along the tooth width. Without considering the contact between two gear teeth, meshing forces are applied on the tooth profile under the plane strain assumption. The inner ring nodes of the gear are coupled with the master node (the geometric centre of the gear), and the master node is restrained from all degrees of freedom. When the deformation in the direction of action line is acquired from FE model and the influence of Hertzian contact is added, the TVMS can be obtained. Likewise, TVMS obtained by FEM under 10 crack cases (see Table 3) are displayed in Fig. 14 and the variation of TVMS is the same as that by method 3.

In this paper, FEM result is considered to be the nearest to the real mesh stiffness. In order to display the differences of three methods clearly, two moments A and B which respectively stand for the initial engagement moment of the cracked tooth and the double-tooth engagement ending moment in a mesh period are selected. Tables 4 and 5 show the mesh stiffness obtained by three methods and the percentage differences of three methods relative to FEM. The bold numbers in Tables 4 and 5 emphasize the method whose result is the nearest to the result of FEM in each crack case.

It can be observed from Table 4 that the results of method 1 are close to the FEM results when the crack depth is smaller, such as 20% and 40% crack, but there is a significant difference when the crack depth is bigger, such as 60% crack. By contrast, the results of methods 2 and 3 are quite close to FEM result. Although the difference becomes pronounced under 60% crack,

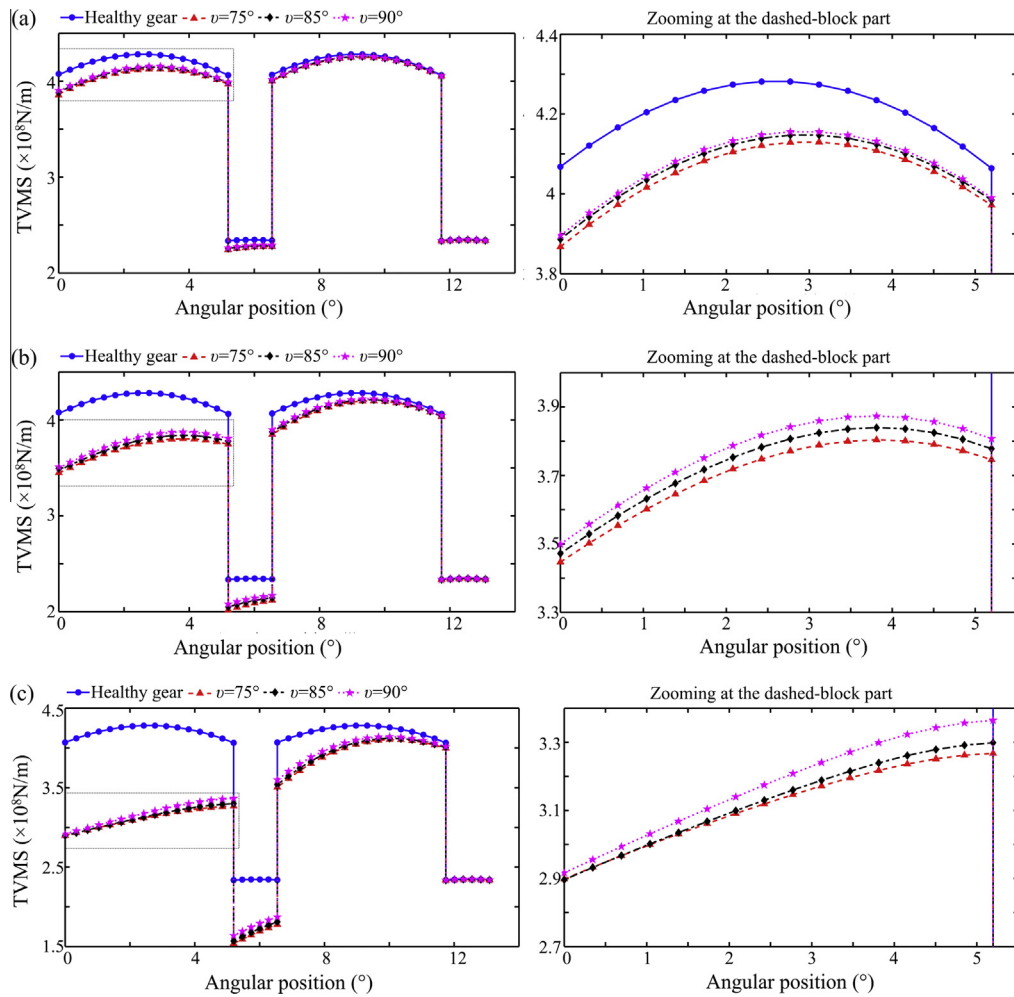


Fig. 10. TVMS of method 2 with different crack depths and propagation directions: (a) 20% crack, (b) 40% crack, (c) 60% crack.

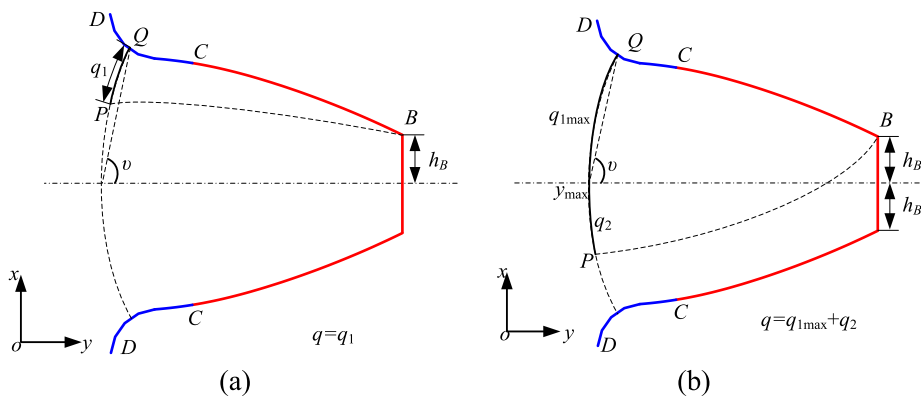


Fig. 11. Schematic of crack propagation in method 3: (a) $q \leq q_{1\max}$, (b) $q > q_{1\max}$.

the difference is almost acceptable. This also shows that the parabolic curve as a limiting line is appropriate. Moreover, in most cases mesh stiffness of method 3 is slightly less than that of method 2 because the tooth has slight rigidity loss compared with straight line assumption in method 2. From Table 5, it can be observed that the differences of three methods are acceptable and the results of methods 2 and 3 are closer to FEM result at moment B.

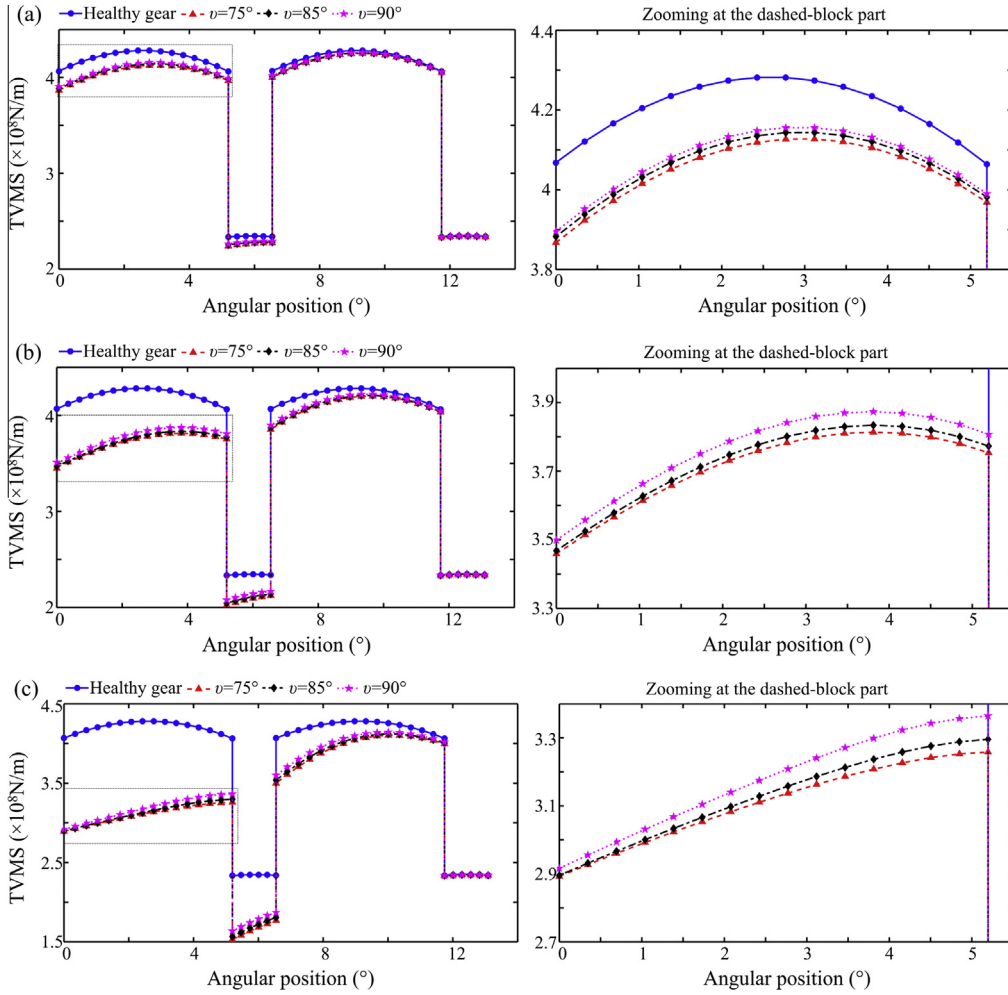


Fig. 12. TVMS of method 3 with different crack depths and propagation directions: (a) 20% crack, (b) 40% crack, (c) 60% crack.

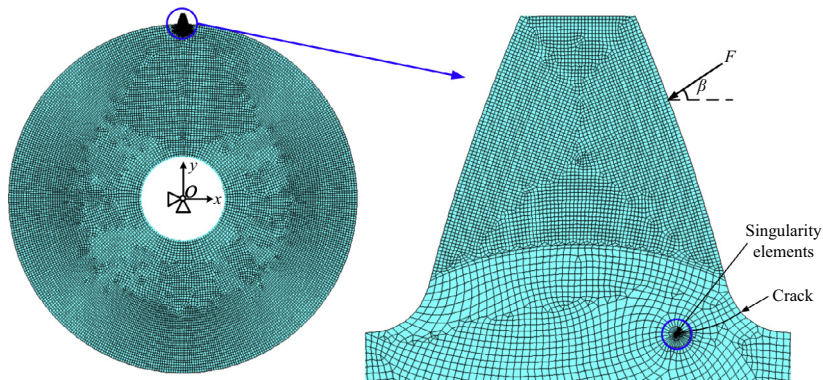


Fig. 13. FE model of a cracked gear.

4. Conclusions

Considering the misalignment of gear root circle and base circle on TVMS, an improved mesh stiffness model for a healthy gear pair is proposed. Three mesh stiffness models for a cracked gear pair are established based on the improved method and compared with FEM. Some conclusions are summarized as follows:

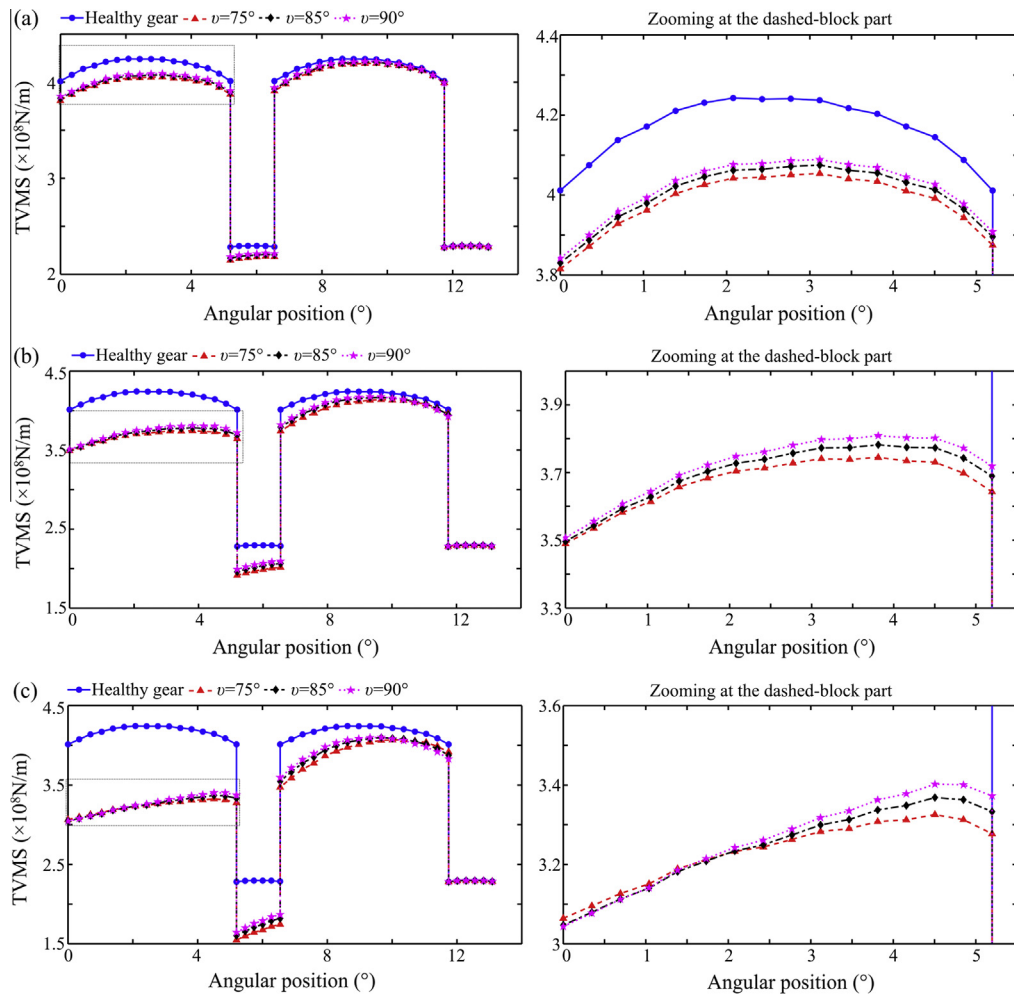


Fig. 14. TVMS of FEM with different crack depths and propagation directions: (a) 20% crack, (b) 40% crack, (c) 60% crack.

Table 4

Percentage differences of three methods compared with FEM at moment A.

Crack depth	v	Method 1		Method 2		Method 3		FEM
		$k_A (\times 10^8 \text{ N/m})$	Difference (%)	$k_A (\times 10^8 \text{ N/m})$	Difference (%)	$k_A (\times 10^8 \text{ N/m})$	Difference (%)	$k_A (\times 10^8 \text{ N/m})$
Healthy		4.068	1.40	4.068	1.40	4.068	1.40	4.012
	75°	3.904	2.31	3.868	1.36	3.868	1.36	3.816
	85°	3.921	2.35	3.887	1.46	3.883	1.36	3.831
	90°	3.929	2.26	3.896	1.41	3.896	1.41	3.842
40%	75°	3.404	−2.46	3.447	−1.23	3.459	−0.89	3.490
	85°	3.432	−1.86	3.472	−0.71	3.468	−0.83	3.497
	90°	3.460	−1.37	3.499	−0.26	3.499	−0.26	3.508
	75°	2.716	−11.36	2.898	−5.42	2.893	−5.58	3.064
60%	85°	2.726	−10.53	2.896	−4.96	2.896	−4.96	3.047
	90°	2.741	−9.95	2.917	−4.17	2.917	−4.17	3.044

Note: k_A stands for the mesh stiffness at moment A.

- (1) The improved mesh stiffness model in which the gear tooth is modeled as a nonuniform cantilever beam on the root circle is more accurate compared with the traditional mesh stiffness model in which the gear tooth is modeled as a cantilever beam on the base circle.
- (2) There is a significant difference between method 1 and FEM under large crack. However, the results of methods 2 and 3 are quite close to FEM result, which shows that the parabolic curve as a limiting line is appropriate.

Table 5

Percentage differences of three methods compared with FEM at moment B.

Crack depth	ν	Method 1		Method 2		Method 3		FEM
		$k_B (\times 10^8 \text{ N/m})$	Difference (%)	$k_B (\times 10^8 \text{ N/m})$	Difference (%)	$k_B (\times 10^8 \text{ N/m})$	Difference (%)	$k_B (\times 10^8 \text{ N/m})$
Healthy		4.064	1.32	4.064	1.32	4.064	1.32	4.011
20%	75°	3.982	2.76	3.972	2.50	3.969	2.43	3.875
	85°	3.993	2.49	3.985	2.28	3.982	2.21	3.896
	90°	3.999	2.30	3.990	2.07	3.990	2.07	3.909
40%	75°	3.738	2.61	3.746	2.83	3.753	3.02	3.643
	85°	3.773	2.28	3.778	2.41	3.773	2.28	3.689
	90°	3.803	2.26	3.807	2.37	3.807	2.37	3.719
60%	75°	3.185	−2.81	3.267	−0.31	3.258	−0.58	3.277
	85°	3.233	−3.00	3.299	−1.02	3.295	−1.14	3.333
	90°	3.303	−2.08	3.365	−0.24	3.365	−0.24	3.373

Note: k_B stands for the mesh stiffness at moment B.

- (3) Mesh stiffness of method 2 is very close to that of method 3. So it is acceptable to assume the crack path to be a straight line to simplify the crack model.

It is worth mentioning that the improved mesh stiffness model proposed in this paper is still powerless when the tooth crack propagates to the foundation of the gear. In future articles, emphasis will be given to the effect of tooth crack on the rigidity of the foundation. In addition, further research is needed during the calculation of gear mesh stiffness considering more realistic crack propagation path, such as the crack propagating to rim.

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Appendix A

$x_\beta, y_\beta, x_1, y_1, x_2, y_2, I_{y1}, I_{y2}, A_{y1}, A_{y2}$ can be expressed as the function of angles γ, β or τ , which can be expressed as:

$$x_\beta = r_b[(\beta + \theta_b) \cos \beta - \sin \beta], y_\beta = r_b[(\beta + \theta_b) \sin \beta + \cos \beta], \quad (\text{A.1})$$

$$x_1 = r \times \sin(\Phi) - (a_1 / \sin \gamma + r_\rho) \times \cos(\gamma - \Phi), y_1 = r \times \cos(\Phi) - (a_1 / \sin \gamma + r_\rho) \times \sin(\gamma - \Phi), \quad (\text{A.2})$$

$$x_2 = r_b[(\tau + \theta_b) \cos \tau - \sin \tau], y_2 = r_b[(\tau + \theta_b) \sin \tau + \cos \tau], \quad (\text{A.3})$$

$$A_{y1} = 2x_1L, A_{y2} = 2x_2L, \quad (\text{A.4})$$

$$I_{y1} = \frac{2}{3}x_1^3L, I_{y2} = \frac{2}{3}x_2^3L, \quad (\text{A.5})$$

So $\frac{dy_1}{d\gamma}$ and $\frac{dy_2}{d\tau}$ can be calculated as follows:

$$\begin{aligned} \frac{dy_1}{d\gamma} = & \frac{a_1 \sin\left(\frac{a_1/\tan\gamma+b_1}{r}\right)(1+\tan^2\gamma)}{\tan^2\gamma} + \frac{a_1 \cos\gamma}{\sin^2\gamma} \sin\left(\gamma - \frac{a_1/\tan\gamma+b_1}{r}\right) \\ & - \left(\frac{a_1}{\sin\gamma} + r_\rho\right) \cos\left(\gamma - \frac{a_1/\tan\gamma+b_1}{r}\right) \left(1 + \frac{a_1(1+\tan^2\gamma)}{r \tan^2\gamma}\right) \end{aligned} \quad (\text{A.6})$$

$$\frac{dy_2}{d\tau} = r_b(\tau + \theta_b) \cos \tau, \quad (\text{A.7})$$

where θ_b is the half tooth angle on the base circle of the gear, $\theta_b = \frac{\pi}{2N} + \text{inv } \alpha$.

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